

PS-13 · AI-Powered Air Quality Forecasting

Delhi AirCast

A simple forecast for your neighbourhood:
PM2.5 and AQI for the next 1 to 24 hours

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Why this project

- People in Delhi need to know the air in their own neighbourhood.
- They also need the next few hours, so they can plan a walk, a run, or school pickup.
- We build Delhi AirCast: a short-term air forecast for each monitoring station.
- The app shows today's air, the next 6–24 hours, and a clear line in simple words.

What the user sees

Right now

Anand Vihar, 10:00

PM2.5 145 $\mu\text{g}/\text{m}^3$

AQI about 187

Category: Poor

Next 6 hours

11 AM 151

12 PM 158

1 PM 163

2 PM 157

4 PM 142

In simple words

Air quality may get worse over the next 3 hours.

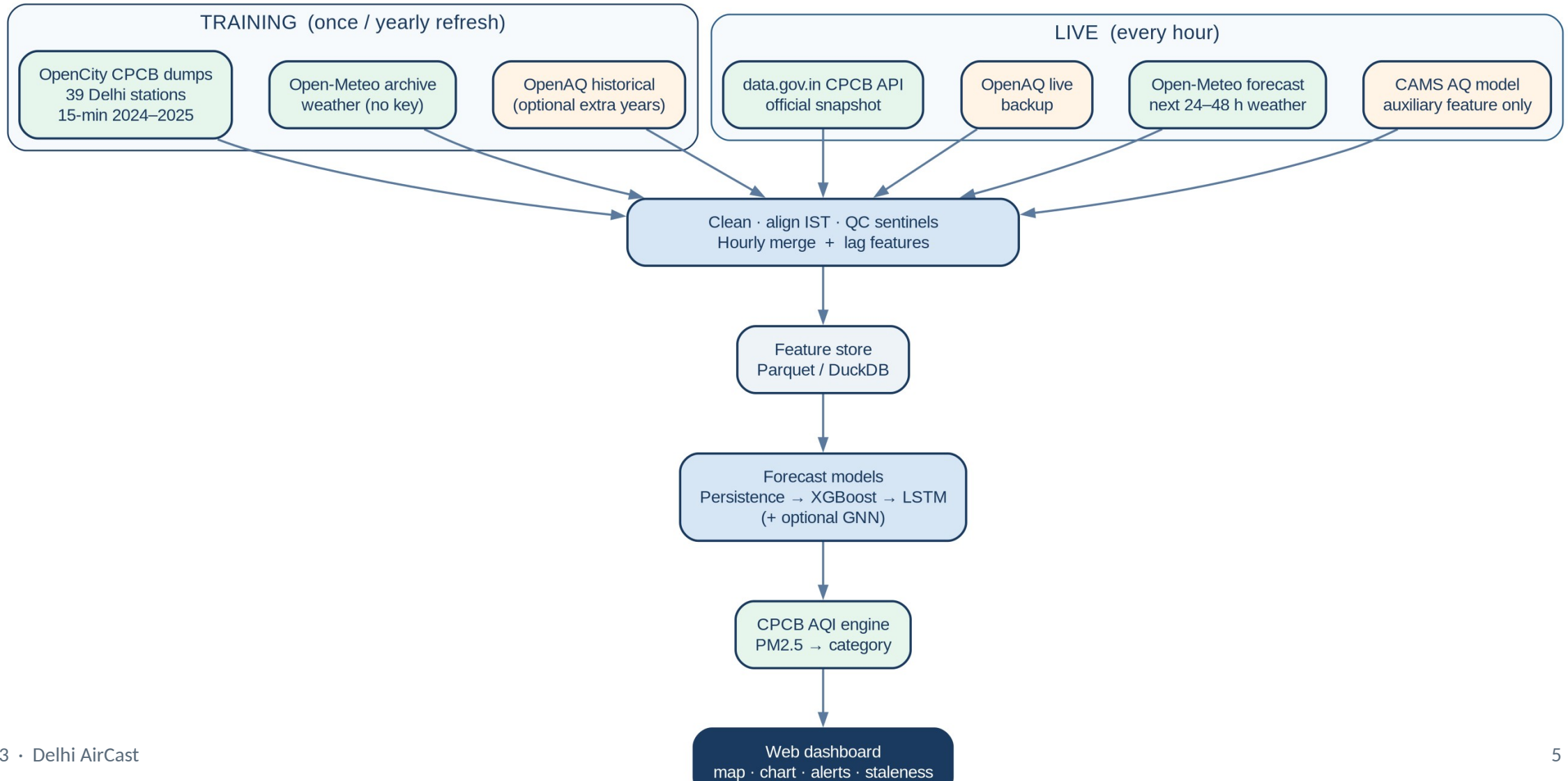
Then it eases a little in the afternoon.

Our data — all free, already working

- Pollution: CPCB / DPCC station data from OpenCity (15-minute files for 2024–2025).
- Live updates: official data.gov.in CPCB API (stations were updating on 7 Sep 2026).
- Weather: Open-Meteo — history and the next 48 hours, no API key.
- Backup: OpenAQ, the same government stations through a second free API.
- We checked these sources ourselves. They respond in a few seconds.

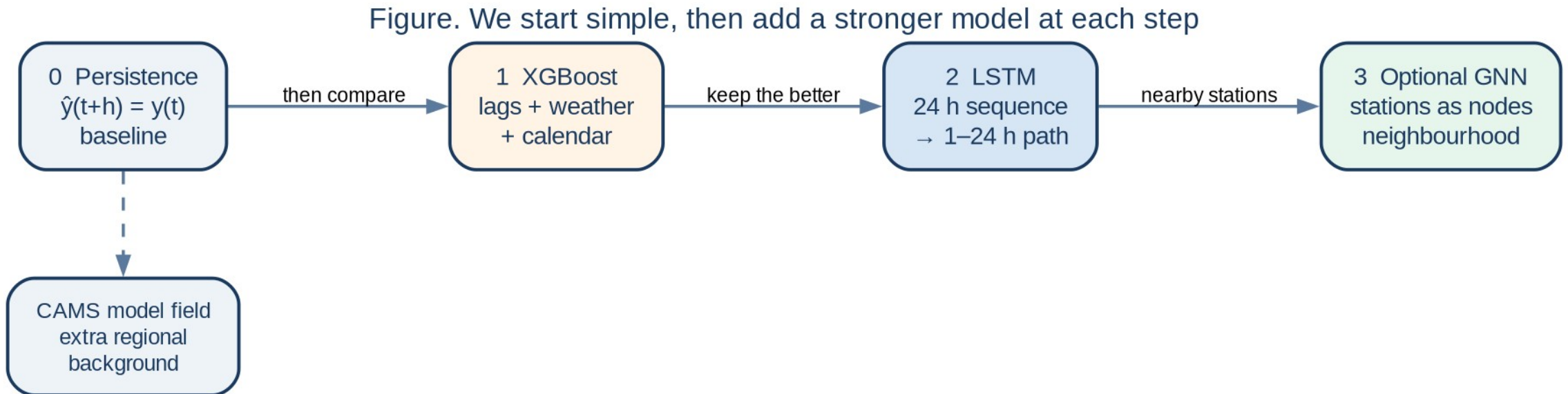
How the system fits together

Figure. End-to-end Delhi AirCast stack (all free sources)

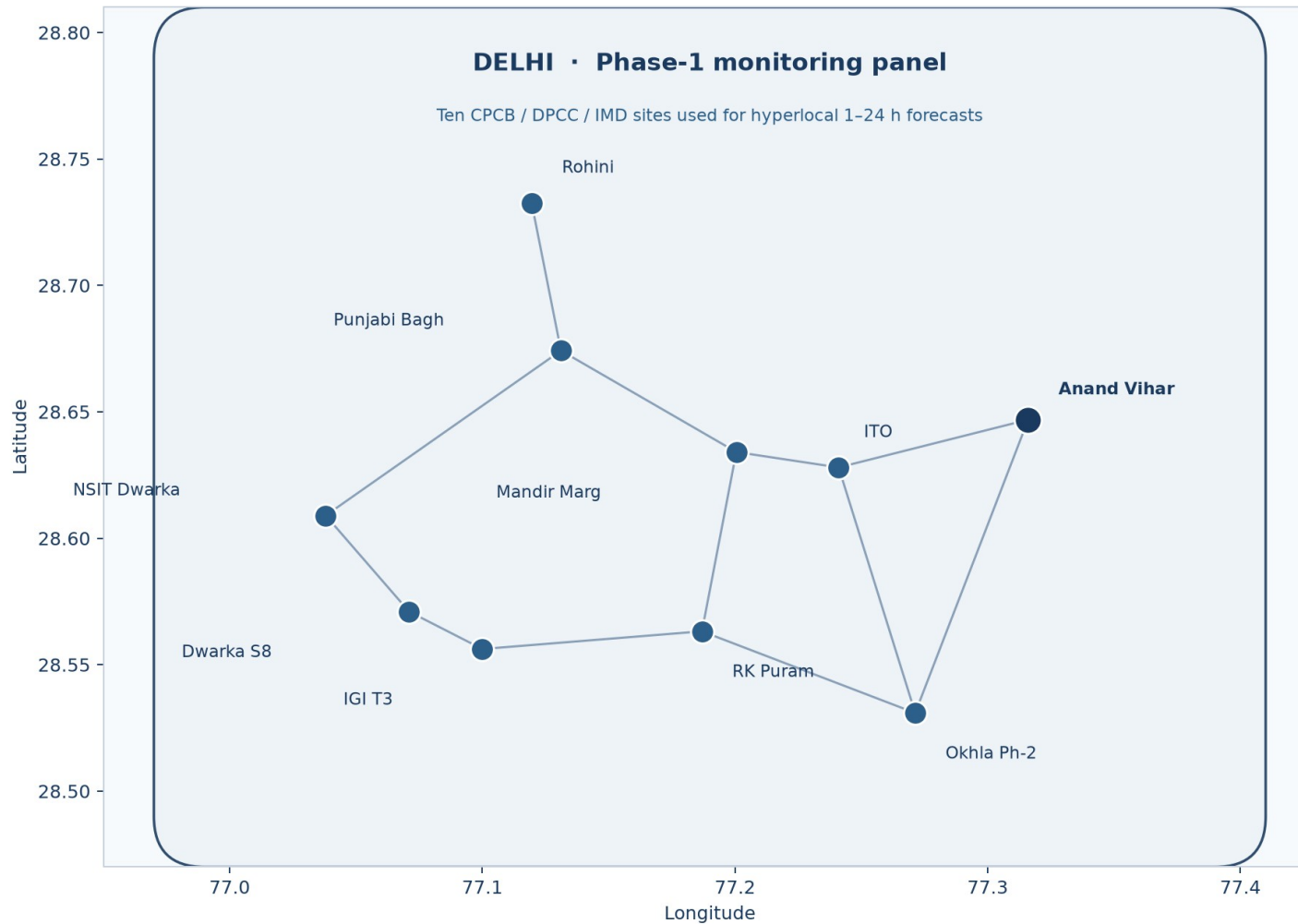


How we forecast

- The model predicts PM2.5. The app then converts that number to Indian AQI.
- We compare three models and keep the one that works best.



Ten stations across Delhi



- Anand Vihar, ITO, Okhla
- Dwarka, NSIT, IGI Airport
- Rohini, Punjabi Bagh
- RK Puram, Mandir Marg
- Each place gets its own forecast.
- Later we can add more stations and nearby NCR cities.

The dashboard

DELHI AIRCAST · Anand Vihar, Delhi — DPCC 10:00 IST

NOW

AQI 187

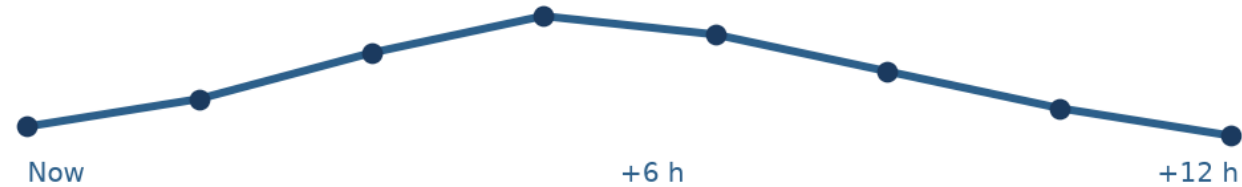
POOR

PM2.5 145 $\mu\text{g}/\text{m}^3$

PM10 220 Temp 27°C Wind 4 km/h

Source: CPCB via data.gov.in

PM2.5 FORECAST (next 12 hours)



Air quality is expected to deteriorate over the next 3 hours.

DELHI SENSOR MAP



SYSTEM HEALTH

- Live feed: CPCB (21:00 IST)
- Weather: Open-Meteo OK
- Backup: OpenAQ standby
- Model: XGBoost v1 (frozen)
- 28-day MAE: tracked continuously
- Retrain: weekly, promote if better

Our plan, and how we grow

1

Clean data

Merge pollution
and weather

2

First models

Simple baseline,
then XGBoost

3

Deeper AI

LSTM, then a
station graph

4

Live app

Hourly update
and weekly improve

- Every hour the app collects new readings and shows a fresh forecast.
- Every week we add the new data and keep the better model.
- The same design can later cover more Delhi stations, NCR, and other cities such as Pune.

Thank you

Delhi AirCast · a clear air forecast for every neighbourhood

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